

True/False (2 points):

6.5



ILLINOIS INSTITUTE OF TECHNOLOGY

Department of Computer Science

CS 458 Final Exam

May 2025

Name: _____

Student ID: A. _____

$$\frac{6.5}{9}$$

$$\frac{6}{10}$$

$$\frac{2}{2}$$

$$\Sigma = 33.5$$

True/False (9 points):

6.5

1. Substitution ciphers work by rearranging the positions of letters in the plaintext. (True / ~~False~~)
2. The Feistel cipher structure is used in both DES and AES. (True / ~~False~~)
3. ECB mode encrypts identical plaintext blocks into identical ciphertext blocks. (~~True~~ / False)
4. The MixColumns operation in AES provides diffusion by mixing the bytes in each column of the state. (~~True~~ / False)
5. Public-key cryptosystems are generally faster than symmetric key systems. (True / ~~False~~)
6. The S-boxes in DES are responsible for the diffusion property of the cipher. (~~True~~ / False)
7. In the Feistel structure, the same algorithm is used for both encryption and decryption. (~~True~~ / False)
8. A major advantage of public-key cryptography is that it eliminates the key distribution and digital signature problems present in symmetric cryptography. (~~True~~ / False)
9. Digital signatures provide non-repudiation, ensuring that the sender cannot deny having sent the message. (True / ~~False~~)
10. The Diffie-Hellman key exchange algorithm can be used for both key exchange and digital signatures. (~~True~~ / False)
11. Digital signatures can provide message confidentiality. (True / ~~False~~)
12. Non-repudiation ensures that the sender of a message can deny the creation of the message. (~~True~~ / False)
13. Digital signatures are more difficult to duplicate or tamper with than conventional signatures. (~~True~~ / False)
14. A cryptographic hash function should be deterministic, meaning the same input data should always produce the same hash output. (True / ~~False~~)
15. It is computationally feasible to find any pairs of inputs $x_1 \neq x_2$ such that $H(x_1) = H(x_2)$ in a secure cryptographic hash function. (True / ~~False~~)
16. The SHA-256 hash function produces a 512-bit hash value. (True / ~~False~~)
17. Cryptographic hash functions are designed to be easily reversible. (True / ~~False~~)
18. Padding schemes in digital signatures are primarily used to prevent attacks such as existential forgery, making it more difficult for attackers to generate valid message-signature pairs. (True / ~~False~~)

Multiple choice (10 points):

1. How many rounds does the AES algorithm use when the key length is 128 bits?
 - ☒ A) 10
 - ☐ B) 12
 - ☐ C) 14
 - ☐ D) 16
2. What does the S-Box in the AES algorithm provide?
 - ☐ A) Key generation
 - ☒ B) Non-linear substitution
 - ☐ C) Permutation
 - ☐ D) Linear transformation
3. The security of RSA is based on the difficulty of which mathematical problem?
 - ☒ A) Discrete logarithm problem
 - ☐ B) Factoring large integers
 - ☐ C) Matrix multiplication
 - ☐ D) Polynomial interpolation
4. What is the main advantage of Elliptic Curve Cryptography (ECC) over RSA?
 - ☐ A) Encryption Algorithm
 - ☒ B) Smaller key size for the same security level
 - ☐ C) Easier key management
 - ☐ D) Simpler implementation
5. In the context of Diffie-Hellman key exchange, what is the primary mathematical problem that ensures its security?
 - ☐ A) Integer factorization problem
 - ☐ B) Discrete logarithm problem
 - ☒ C) Elliptic curve problem
 - ☐ D) RSA problem
6. Which of the following is not a core security service provided by digital signatures?
 - ☐ A) Message Integrity
 - ☒ B) Message Confidentiality
 - ☐ C) Message Authentication
 - ☐ D) Non-repudiation

7. What cryptographic method fails to provide a desirable security function in the example of Bob ordering a pink car?

- A) Symmetric cryptography
- B) Asymmetric cryptography
- ☒ C) Public Key Infrastructure (PKI)
- D) Digital certificates

8. What ensures that a message has not been modified in transit?

- A) Confidentiality
- ☒ B) Message Integrity
- C) Non-repudiation
- D) Authentication

9. Which digital signature scheme is known for verifying a message by decrypting the signature with the signer's public key?

- ☒ A) ElGamal Digital Signature Scheme
- B) Digital Signature Standard (DSS)
- C) Schnorr Digital Signature Scheme
- D) RSA Digital Signature Scheme

10. Which property ensures that it is infeasible to find two different inputs that produce the same hash output?

- a) Pre-image resistance
- ☒ b) Collision resistance
- c) Second pre-image resistance
- d) Deterministic property

Break The RSA and produce Digital Signature (21 points):

We have an RSA public-key cryptosystem between Alice and Bob.

Alice's public key: $(N=33, e=3)$

Bob's Public key: $(N=21, e=5)$

1. (7 points) Break the system and find $P, q, \phi(N)$, private key d . For both Alice and Bob (Show your work in detail and write all formulas)

Alice: $p=3, q=11$
 $\phi(N) = (3-1)(11-1) = 20 = \phi(N)$
 $3 \times d_s \bmod 20 = 1$
 $d_s = 7$

Bob: $p=3, q=7$
 $\phi(N) = (3-1)(7-1) = 12 = \phi(N)$
 $5 \times d_r \bmod 12 = 1$
 $d_r = 5$

$2 \times \frac{25}{12} = 24 \frac{1}{6}$

2. (14 points) Bob wants to sign a message ($M = 4$), then encrypt it and send it to Alice. Explain the whole process and calculations, and find all the numbers, such as: (Show your work in detail and write all formulas)

Bob's tasks:

Bob as the sender uses the Message (4), signs it w/ his priv key and uses his N .

- a. Sign a message ($M = 4$) and calculate the signature value ($S = \text{Signature Value}$)

$$S = M^{d_s} \bmod N_s$$

$$= 4^5 \bmod 21$$

$$S = 16$$

$$\begin{array}{r} 1024 \\ 21 \overline{) 40} = 19 \\ \underline{39} \\ 1 \\ 5 \\ \underline{3} \\ 2 \end{array}$$

$$\begin{array}{r} 1024 \\ \underline{184} \\ 840 \\ \underline{105} \\ 74 \\ \underline{63} \\ 16 \end{array}$$

encrypt it using alices e , and her N .

- b. Encrypt signature value and send it to Alice ($C = \text{Encrypted Signature Value}$)

$$C = S^e \bmod N_A$$

$$C = 16^3 \bmod 33$$

$$C = 4$$

$$16 \times 16 = 256$$

$$256 \times 16 =$$

$$256 \times 10 = 2560$$

$$256 \times 2 = 512$$

$$512$$

$$512$$

$$1536$$

$$2560$$

$$4096$$

$$- 3300 \quad 33 \quad 100$$

$$796$$

$$- 660 \quad 20$$

$$136$$

$$132 \quad 4$$

$$4$$

Alice's tasks:

Alice uses her priv key to decrypt the msg

c. Decrypt the received encrypted signature ($S' = \text{Decrypted Signature Value}$)

$$S' = C^{d_A} \bmod N_A$$

$$= 4^7 \bmod 33$$

$$S' = 16$$

$$4 \times 4 = 16$$

$$\times 4 = 64$$

$$\times 4 = 256$$

$$\times 4 = 512$$

$$\times 4 = 2048$$

$$\times 4 = 8196$$

$$\begin{array}{r} 16394 \\ - 9900 \\ \hline 6494 \\ - 3300 \\ \hline 3194 \\ - 2640 \\ \hline 554 \\ - 330 \\ \hline 224 \\ - 16 \\ \hline 208 \end{array}$$

Use S' with sender's encrypt + N to get M'

d. Verify the signature and signer identity ($M' = \text{Extracted Message Value}$)

$$M' = S'^e \bmod N_S$$

$$M' = 16^5 \bmod 21$$

$$16 \times 16 = 256$$

$$16 \times 6 = 96$$

$$= 256 \bmod 21$$

$$- 210$$

$$46$$

$$- 42$$

$$4$$

$$M' = 4$$

$$M' = M = 4$$

True the signature is
verified and the identity
is Bob

1. True/False (20 points):

1. Replay attack (True/False)
2. Substitution ciphers work by rearranging the positions of letters in the plaintext. (True/False)
3. In a Vigenere cipher, a single key is applied to each letter of the message. (True/False)
4. The Feistel cipher structure is used in both DES and AES. (True/False)
5. The primary weakness of DES is its small key size. (True/False)
6. ECB mode encrypts identical plaintext blocks into identical ciphertext blocks. (True/False)
7. Information assurance involves protecting the confidentiality, integrity, and availability of information. (True/False)
8. The MixColumns operation in AES provides diffusion by mixing the bytes in each column of the state. (True/False)
9. The initial and final permutations in DES have significant cryptographic importance. (True/False)
10. Public-key cryptosystems are generally faster than symmetric key systems. (True/False)
11. In symmetric cryptography, a stream cipher encrypts data in fixed-size blocks. (True/False)
12. The S-boxes in DES are responsible for the diffusion property of the cipher. (True/False)

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13. The stream cipher uses a single key for both encryption and decryption. (True/False)
14. In the Feistel cipher, the same algorithm is used for both encryption and decryption. (True/False)
15. The avalanche effect ensures that a small change in plaintext results in a significant change in ciphertext. (True/False)
16. A major advantage of public-key cryptography is that it eliminates the key distribution and digital signature problems present in symmetric cryptography. (True/False)
17. The security of RSA is based on the difficulty of factoring large integers. (True/False)
18. Digital signatures provide non-repudiation, ensuring that the sender cannot deny having sent the message. (True/False)
19. Padding is used in RSA to prevent deterministic encryption. (True/False)
20. The Diffie-Hellman key exchange algorithm can be used for both key exchange and digital signatures. (True/False)

Grade (out of 62): 29

1. True/False (20 points):

12

1. Replay attacks are a type of passive attack. (True / False)
2. Substitution ciphers work by rearranging the positions of letters in the plaintext. (True / False)
3. In a Vigenère cipher, the same substitution is applied to each letter of the message. (True / False)
4. The Feistel cipher structure is used in both DES and AES. (True / False)
5. The primary weakness of DES is its small key size. (True / False)
6. ECB mode encrypts identical plaintext blocks into identical ciphertext blocks. (True / False)
7. Information assurance involves protecting the confidentiality, integrity, and availability of information. (True / False)
8. The MixColumns operation in AES provides diffusion by mixing the bytes in each column of the state. (True / False)
9. The initial and final permutations in DES have significant cryptographic importance. (True / False)
10. Public-key cryptosystems are generally faster than symmetric key systems. (True / False)
11. In symmetric cryptography, a stream cipher encrypts data in fixed-size blocks. (True / False)
12. The S-boxes in DES are responsible for the diffusion property of the cipher. (True / False)
13. The Counter (CTR) mode of operation allows block ciphers to function like stream ciphers. (True / False)
14. In the Feistel structure, the same algorithm is used for both encryption and decryption. (True / False)
15. The Avalanche effect in DES ensures that a small change in plaintext results in a significant change in ciphertext. (True / False)
16. A major advantage of public-key cryptography is that it eliminates the key distribution and digital signature problems present in symmetric cryptography. (True / False)
17. The security of RSA is based on the difficulty of factoring large integers. (True / False)
18. Digital signatures provide non-repudiation, ensuring that the sender cannot deny having sent the message. (True / False)
19. Padding is used in RSA to prevent deterministic encryption. (True / False)
20. The Diffie-Hellman key exchange algorithm can be used for both key exchange and digital signatures. (True / False)

2. Multiple choice (15 points):

1. Which principle states that a cryptographic system should remain secure even if everything about the system, except the key, is public knowledge?
 - • ☒ A) Kerckhoffs' Principle
 - B) Shannon's Principle
 - C) Caesar's Principle
 - D) Diffusion Principle
2. In a symmetric cryptosystem, what is true about the keys used for encryption and decryption?
 - • ☒ B) Only one key is used for both encryption and decryption
 - A) They are different
 - C) They are always public
 - D) They are generated randomly each time
3. What type of cipher is the Caesar cipher?
 - • ☒ C) Mono-alphabetic substitution cipher
 - A) Transposition cipher
 - B) Polyalphabetic cipher
 - D) Stream cipher
4. Which of the following algorithms is used for key exchange but not for encryption?
 - • ☒ C) Diffie-Hellman
 - A) RSA
 - B) AES
 - D) ElGamal
5. What is the main advantage of using the one-time pad cipher?
 - • ☒ B) It is provably secure if used correctly
 - A) It is very easy to use
 - C) It uses very short keys
 - D) It requires no key management
6. Which of the following best describes the concept of 'diffusion' in cryptography?
 - • ☒ B) Spreading the influence of each plaintext element over many ciphertext elements
 - A) Making the relationship between the ciphertext and the key as complex as possible
 - C) Encrypting each character of the plaintext with a different key
 - D) Ensuring that the same key is not used more than once

7. Which type of cipher rearranges the characters of the plaintext without changing the actual characters used?

- A) Substitution cipher
- • ☒ B) Transposition cipher
- C) Stream cipher
- D) Block cipher

8. In the context of cryptography, what is 'confusion'?

- A) The process of spreading the plaintext over the ciphertext
- • ~~B) The complexity of the relationship between the plaintext and the ciphertext~~
- • ☒ C) The mixing of multiple ciphers
- D) The secret transmission of the key

9. In the context of symmetric encryption, what is a stream cipher?

- A) A cipher that encrypts data in fixed-size blocks
- • ☒ B) A cipher that encrypts data one bit or one byte at a time
- C) A cipher that uses asymmetric keys
- D) A cipher that uses multiple encryption rounds

10. What is the block size used by the AES algorithm?

- A) 64 bits
- • ☒ B) 128 bits
- C) 192 bits
- D) 256 bits

11. How many rounds does the AES algorithm use when the key length is 128 bits?

- • ~~A) 10~~
- B) 12
- C) 14
- • ☒ D) 16

$$\begin{array}{l} 6 \times 2 = 120 \\ 6 \times 3 = 48 \\ 6 \times 100 = 600 \\ \hline 768 + 1280 \end{array}$$

12. What does the S-Box in the AES algorithm provide?

- A) Key generation
- • ☒ B) Non-linear substitution
- C) Permutation
- D) Linear transformation

13. The security of RSA is based on the difficulty of which mathematical problem?

- A) Discrete logarithm problem
- • ☒ B) Factoring large integers
- C) Matrix multiplication
- D) Polynomial interpolation

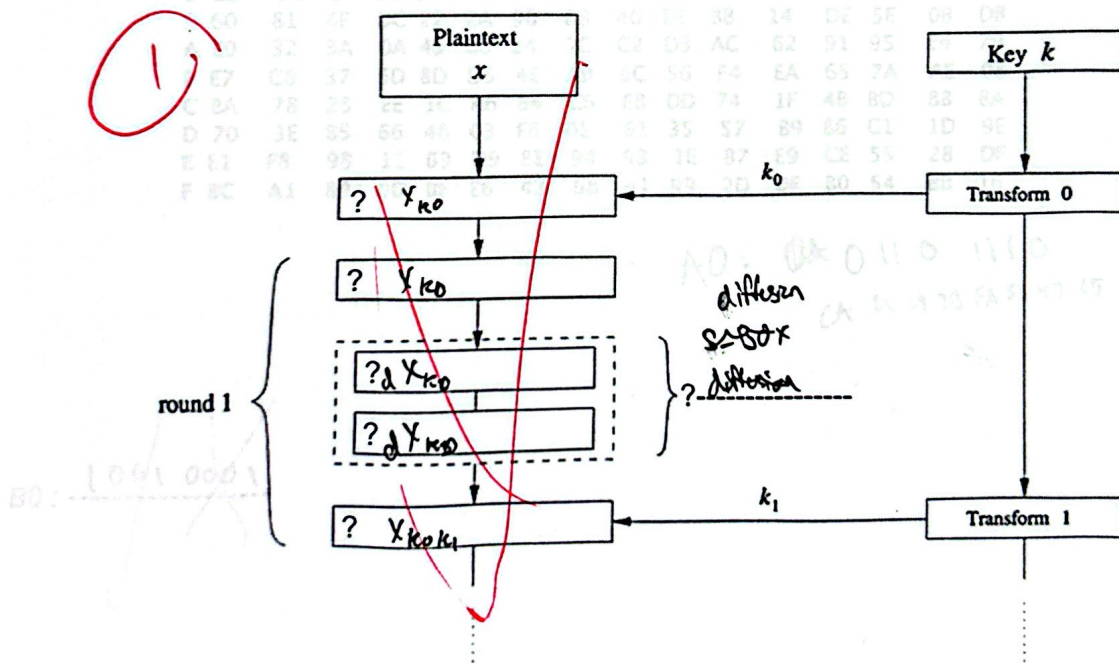
14. What is the main advantage of Elliptic Curve Cryptography (ECC) over RSA?

- A) Encryption Algorithm
- • B) Smaller key size for the same security level
- C) Easier key management
- D) Simpler implementation

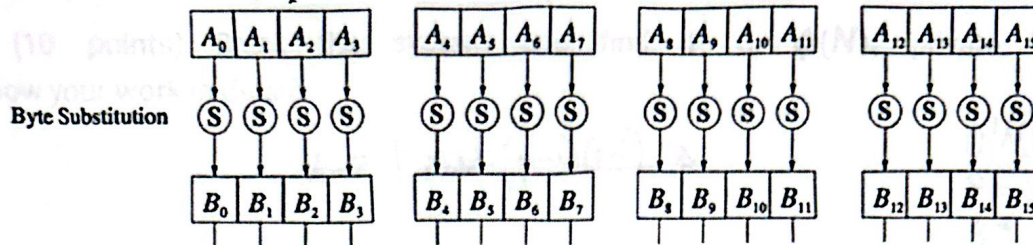
15. In the context of Diffie-Hellman key exchange, what is the primary mathematical problem that ensures its security?

- **A) Integer factorization problem**
- B) Discrete logarithm problem
- C) Elliptic curve problem
- D) RSA problem

3. AES Structure: Fill in the six blank boxes marked with question marks (6 points)



4. AES S-Box: Substitution values in hexadecimal notation for input byte (xy). (6 points)



Use the given AES S-box table to calculate B0 if A0 : 01101110 (Show your work in detail and give B0 in binary format like A0)

0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0 63	7C	77	7B	F2	6B	6F	C5	30	01	67	2B	FE	D7	AB	76
1 CA	82	C9	7D	FA	59	47	F0	AD	D4	A2	AF	9C	A4	72	C0
2 87	FD	93	26	36	3F	F7	CC	34	A5	E5	F1	71	D8	31	15
3 04	C7	23	C3	18	96	05	9A	07	12	80	E2	EB	27	B2	75
4 09	83	2C	1A	1B	6E	5A	A0	52	38	D6	83	29	E3	2F	84
5 53	D1	00	ED	20	FC	B1	5B	6A	CB	BE	39	4A	4C	58	CF
6 D0	EF	AA	FB	43	4D	33	85	45	F9	02	7F	50	3C	9F	A8
7 51	A3	40	8F	92	9D	38	F5	BC	B6	DA	21	10	FF	F3	D2
8 CD	0C	3	EC	5F	97	44	17	C4	A7	7E	3D	64	5D	19	73
9 60	81	4F	DC	22	2A	90	88	46	EE	B8	14	DE	5E	0B	DB
A E0	32	3A	0A	49	06	24	5C	C2	D3	AC	62	91	95	E4	79
B E7	C8	37	6D	8D	D5	4E	A9	6C	56	F4	EA	65	7A	AE	08
C BA	78	25	2E	1C	A6	B4	C6	E8	DD	74	1F	4B	BD	8B	8A
D 70	3E	B5	66	48	03	F6	0E	61	35	57	B9	86	C1	1D	9E
E E1	F8	98	11	69	D9	8E	94	9B	1E	87	E9	CE	55	28	DF
F 8C	A1	89	0D	BF	E6	42	68	41	99	2D	0F	B0	54	BB	16

A0 = 01101110
 CA 82 49 7D FA 51 47 15

B0: 10010001

5. Break The RSA (15 points):

We have an RSA public-key cryptosystem with a public key ($N=39$, $e=5$).

1. (10 points) Break the system and find: P , q , $\phi(N)$, private key d .
(Show your work in detail)

$$d = \text{mod}(p-1)(q-1)$$

$$\begin{aligned}\phi(N) &= 39^5 \\ p &= 13 \\ q &= 3 \\ d &= 38\end{aligned}$$

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2. (5 points) We have a cipher-text: $C=3$. Recover corresponding plain-text M from this cipher-text using your calculated private key. (Show your work)

$$M = 3 \text{ mod } 38$$

$$M = 2$$

$$3 \times 12 = \frac{36}{2}$$

f